AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

# Particulate Matter Monitoring & Health Effects

Issue 2026-09-08 | Entry window 8 September 2026 | 18 records screened in scope

**18** RECORDS IN SCOPE (34 EXCLUDED)

**7** SHIPPED ON METADATA ONLY

**4** WITH A HUMAN HEALTH ENDPOINT

**12** ON THE SENSING / EXPOSURE AXIS

**13** NOT VIA THE PUBMED LEGS

## Signal of the day

**A third of the variance in indoor $PM_{2.5}$ sits *between* homes and survives adjustment for outdoor concentration, windows, cooking and purifier use — which is the quantitative case for per-dwelling measurement rather than per-city inference.**

Forty-eight residences across eight Northern Chinese cities, paired indoor and outdoor $PM_{2.5}$ at 5-minute resolution over seven consecutive days in each of four seasons, aggregated to **31,094 valid hourly observations** of 32,256 scheduled (*Atmosphere*). The linear mixed-effects model carries a residence-level random intercept and an AR(1) residual structure, and the two $R^2$ values are the finding: **marginal** $R^2 = 0.71$ for the fixed effects alone against **conditional** $R^2 = 0.91$ once the dwelling identity is admitted, with **ICC** $= 0.34$. The fixed effects behave as expected — outdoor $PM_{2.5}$ $\beta = 0.872$, window-open fraction 0.548, cooking 0.046, purifier-on fraction $-0.902$, with continuous purifier operation associated with **59.4% lower** indoor $PM_{2.5}$ — and the authors are careful to call that last figure observational rather than causal. What matters for network design is the residual third: after the outdoor field, the occupant behaviour and the building covariates are all in the model, dwelling identity still carries a third of the variance, and no ambient monitor or infiltration-factor lookup recovers it. Eight cities of Northern China with coal-based district heating is a specific and unusually dirty regime, seven days per season is short for a behavioural covariate, and window-open fraction and purifier use are self-reported into the same model that they dominate.

## What changed today

- **Seven of eighteen records ship with no abstract in existence anywhere.** This is new, and it is a property of same-day windowing rather than of the papers. Nine Crossref-by-ISSN records carried no abstract in the journal-scoped payload; each was then re-checked *per DOI* against Crossref directly, against Europe PMC, against Semantic Scholar and against the publisher page. All four legs returned nothing — ScienceDirect and Taylor & Francis both refused the fetch outright. Seven are unambiguously in scope on the title and ship as `Metadata only`, tier C, with the finding fields stating what is and is not established; two more were rejected because without an abstract their scope cannot be confirmed as particulate at all. **Crossref indexes a DOI faster than publishers deposit the abstract, and a one-day entry window sits inside that gap.**
- **Crossref-by-ISSN supplied 12 of 18 records; the PubMed spine supplied 5.** The by-ISSN leg returned 59 raw and shipped 12 — *Aerosol Sci Technol*, *J Aerosol Sci* twice, *Atmos Meas Tech*, *Atmos Chem Phys*, *Atmosphere* three times, *Atmos Pollut Res*, *Indoor Environments*. Nine of those twelve sit in journals the PubMed spine does not index at all. Two more (*Environ Int*, *Environ Pollut*) *are* PubMed-indexed and still arrived here first, which is the same deposit-lag asymmetry seen from the abstract side.
- **The PubMed connector added nothing: the harvester's recall was complete.** Run independently on both axes with an `[EDAT]` window, the connector returned **19 health-axis** PMIDs — the identical set the harvester fetched — and **2 sensing-axis**, a subset of the harvester's 3. First run in this archive where the connector is a pure confirmation rather than a supplement.
- **Consensus returned three hits and all three were already published here** — 1, 2 and 5 August. Verified by resolving each title to its DOI and looking it up in `seen.json`, not by inspecting publication dates, which closes the verification weakness noted on 4 September. The dedup index did exactly what it exists to do.
- **The trial-watch zero is topical, not a registry outage.** ClinicalTrials.gov posted **1,135 updates registry-wide** on 8 September and **0** matching air pollution or particulate matter. Yesterday's zero was a Labor Day artefact and could not distinguish the two cases; today's no-term control separates them cleanly. `trials.json` unchanged at 11.
- **Every confidence interval in the issue comes from one paper.** The forest plot holds four pooled estimates from a single systematic review, and its own heterogeneity statistics are the reason to read it sceptically: $PM_{10}$ gives the largest risk ratio, **1.27 (1.11–1.47)**, on **three studies at** $I^2 = 99.5\%$; $PM_1$ gives the only heterogeneity-free estimate, **1.07 (1.05–1.09)**, $I^2 = 0.0\%$, on **two**. The review's title promises polypharmacy and it found **zero eligible studies** on it, and none on household air pollution, older adults, or the Americas and Oceania.

- **Two source legs were rate-limited rather than broken.** arXiv returned HTTP 429 where it has previously returned an unusable date filter, and OpenAlex returned 429 rather than its usual 403 paywall message. Neither contributed. The OpenAlex leg remains broken for a different reason and is still routed around.

## Corpus analytics

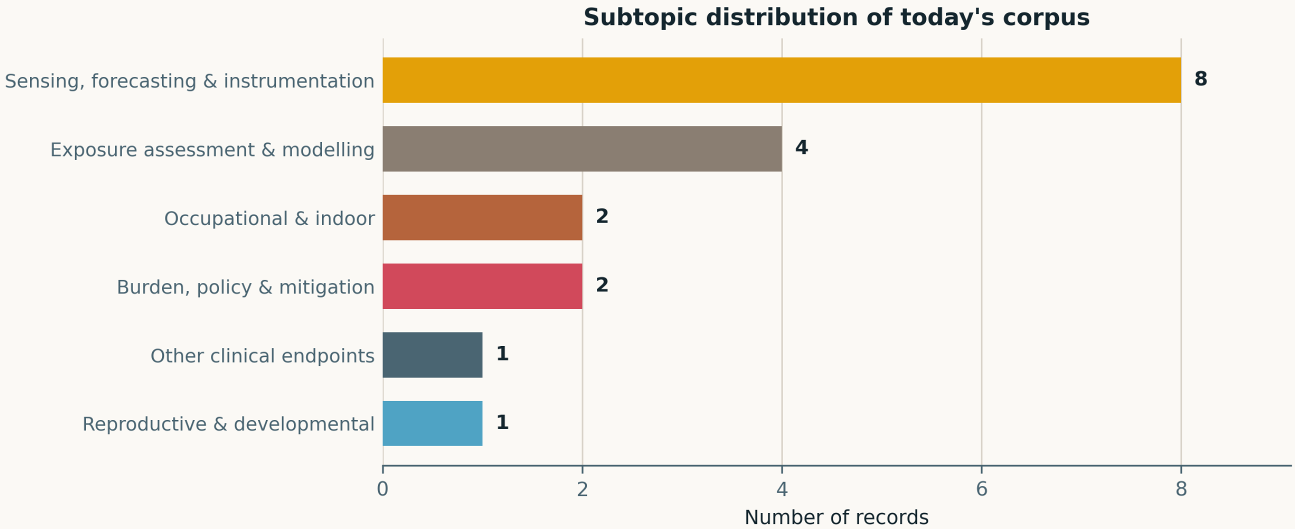
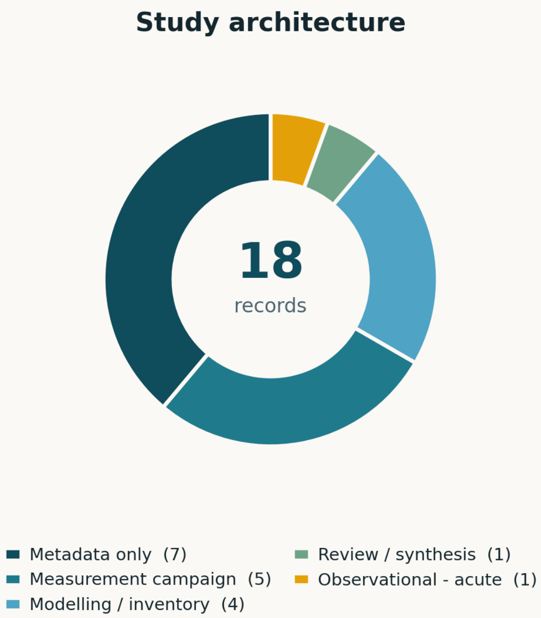



**Fig.** Subtopic assignment of the 18 in-scope records. Only **six of the ten clusters are occupied**, the narrowest spread in recent issues: *Mechanistic toxicology*, *Cardiovascular & metabolic*, *Neuro / mental health* and *Respiratory & allergic* are all empty. *Sensing, forecasting & instrumentation* takes **8 of 18** on its own. This is a source-leg effect — Crossref-by-ISSN watches instrumentation journals and fired hard today — not a shift in the field.



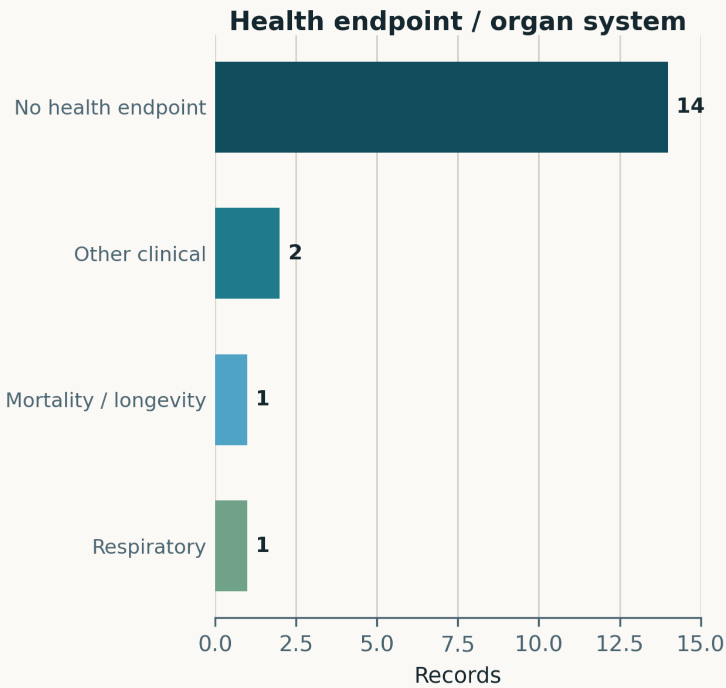




**Fig.** Left — study architecture. The legend sums to 18 as **Metadata only 7, measurement campaign 5, modelling / inventory 4, review / synthesis 1, observational (acute) 1**. The *Metadata only* slice is the largest and is an artefact of the entry window, not a description of the papers; every record in it is a real study whose abstract had not deposited by the time the window closed. **Exactly one observational epidemiological design appears in the whole issue.** Right — health endpoint. **Fourteen of 18 carry no health endpoint**, and all four that do are human, so the human-endpoint count is 4.

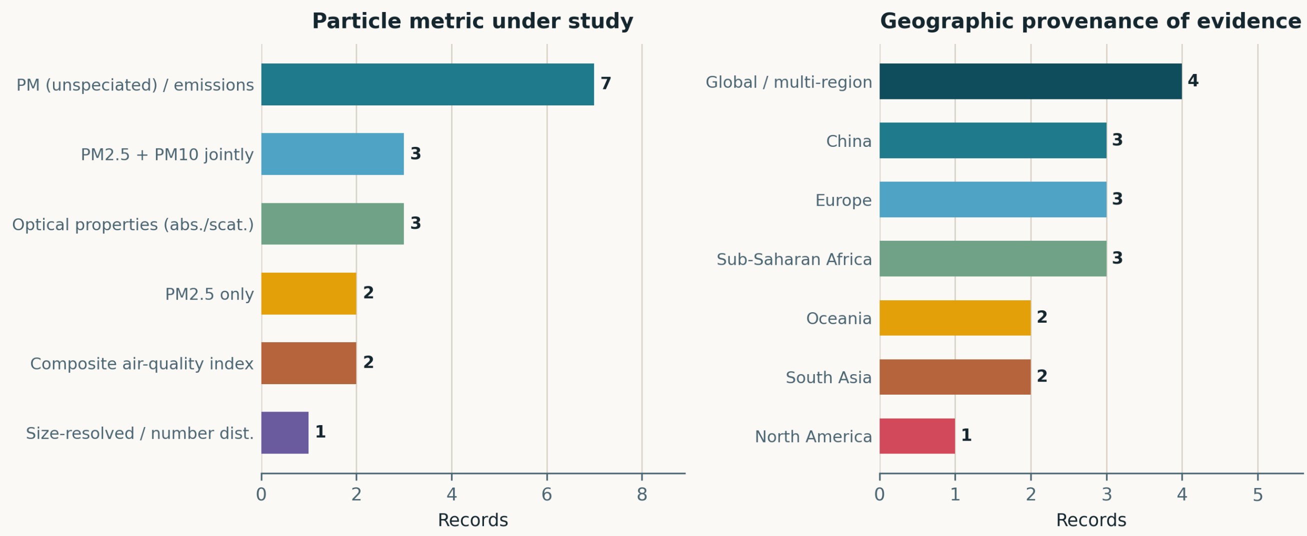



**Fig.** Left — particle metric. *PM (unspeciated) / emissions* is the largest bin at 7 of 18; only **one record reports a size-resolved or number metric** (Aitken-mode particle number, and that one on metadata alone). *Optical properties* at 3 is unusually high and reflects the instrumentation skew. Right — geographic provenance. No real region dominates: Sub-Saharan Africa, Europe and China tie at 3 each. Three Sub-Saharan records — Accra, Ethiopia and a Western Sub-Saharan regional review — ties the archive's daily high, set on 6 August and matched on 3 September, but at **3 of 18 this is the largest *share* the archive has recorded** (16.7%, against 11.1% and 7.9%). The nominally largest bar, *Global / multi-region* at 4, is a fallback: one genuine multi-region meta-analysis, one bench study, and two metadata-only records with no site stated.

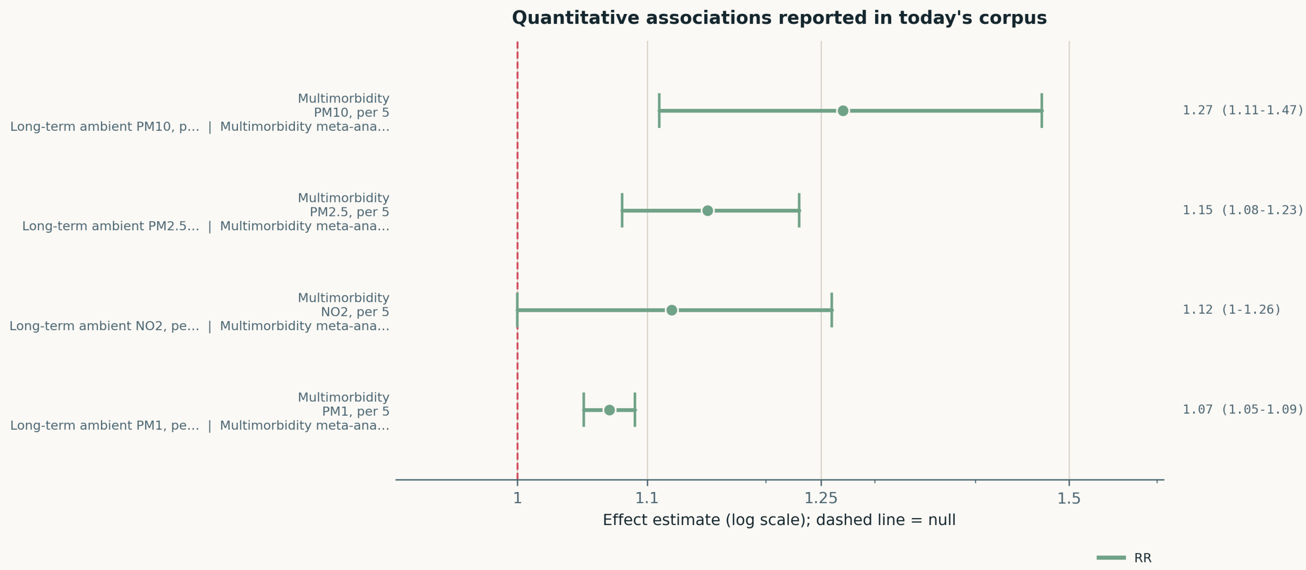



**Fig.** All four quantitative estimates in the corpus, on a common logarithmic axis. **They are four pooled estimates from one systematic review, not four studies**, so the panel shows a single meta-analysis's internal contrasts and nothing about between-study agreement. Each is a random-effects adjusted risk ratio for multimorbidity per 5 $\mu g/m^3$. Read the heterogeneity with them: $I^2$ is **93.0%** for $PM_{2.5}$ (4 studies), **99.5%** for $PM_{10}$ (3), **0.0%** for $PM_1$ (2) and **96.6%** for $NO_2$ (4). The $NO_2$ interval touches 1.00. The visually widest estimate, $PM_{10}$, is also the least interpretable.

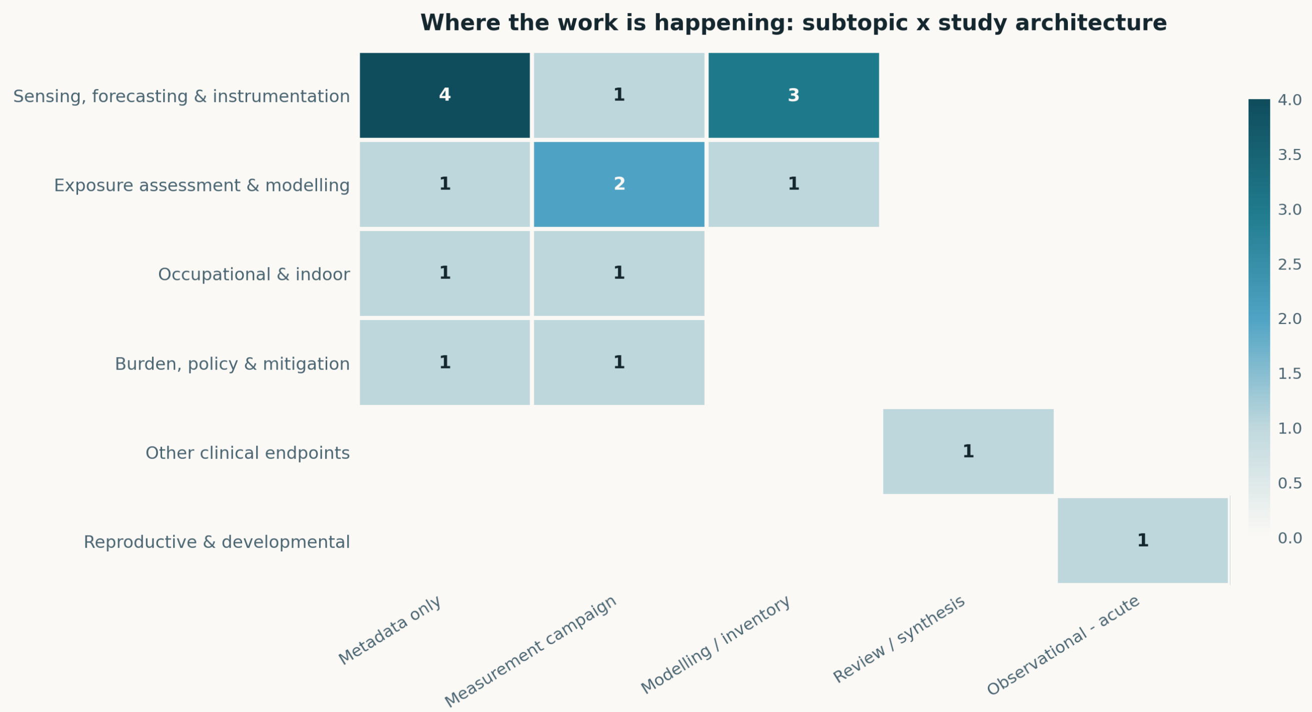



**Fig.** Subtopic × study-architecture cross-tabulation. The single densest cell is *Sensing* × *Metadata only* at **4 records**, which is a statement about publisher deposit latency in instrumentation journals rather than about study design. Twelve cells are occupied out of sixty. The structural absence is the same one this watch keeps recording: **no cell links an observational design to the sensing or exposure clusters**. Nothing in this window measures particles with a novel or low-cost instrument *and* attaches a health outcome to what it measured; the Oklahoma cultivation-facility study comes closest and stops at exposure characterisation.

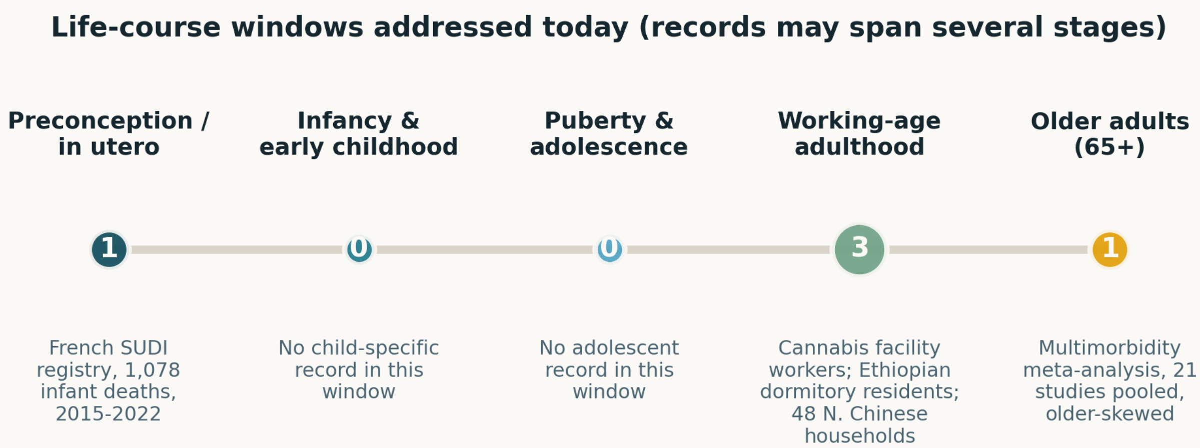



**Fig.** Life-course windows addressed; counts are non-exclusive and only **5 of the 18 records** place anywhere on this axis. Three of the five carry a human health endpoint; the other two — the Ethiopian dormitories and the 48 Northern Chinese homes — place by the population whose exposure is characterised rather than by an outcome.

# Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Entries marked METADATA ONLY had no abstract in existence on the entry date; for these the entry states what the record establishes and what it does not, and asserts no finding.

## Sensing, forecasting & instrumentation — 8 records

**N China indoor Finf 48 homes 2026** Atmosphere

*Determinants of indoor $PM_{2.5}$ and infiltration factors across 48 dwellings*

Paired indoor/outdoor $PM_{2.5}$ at 5 min in 48 residences across eight Northern Chinese cities, seven consecutive days in each of four seasons, **31,094 valid hourly observations** from 32,256 scheduled. A linear mixed-effects model with a residence random intercept and AR(1) residuals gives **marginal** $R^2 = 0.71$**, conditional** $R^2 = 0.91$**, ICC** $= 0.34$: dwelling identity carries a third of the variance that the fixed effects — outdoor $PM_{2.5}$ ($\beta = 0.872$), window-open fraction ($0.548$), cooking ($0.046$), purifier-on fraction ($-0.902$) — cannot reach. Continuous purifier operation associates with **59.4% lower** indoor $PM_{2.5}$, which the authors correctly decline to call causal. Threat to inference: window and purifier use are self-reported and dominate the same model, and coal-heated Northern China is a specific regime. For distributed sensing this is the cleanest recent argument that per-dwelling measurement is not redundant with a good ambient field.

doi: 10.3390/atmos17090874

**Europe BLH regression kriging 2026** Atmos Meas Tech

*Monthly boundary-layer height maps for Europe at 25 km from radiosonde, satellite and ERA5*

Regression kriging over 2010–2020 combining radiosonde profiles with land-surface temperature, topographic covariates and ERA5, at 25 km and monthly means for 00 and 12 UTC. Random forest beat linear regression and gradient boosting at both hours. Against ERA5 the product is consistently sharper: winter spatial variability at 12 UTC gives **RMSE** $\leq$**170 m versus** $\geq$**250 m**, summer temporal variability $\leq$**260 m versus 340 m**. Performance **declines considerably at 00 UTC**, which is the honest limitation and the awkward one — the nocturnal stable layer is exactly where mixing-height error propagates hardest into surface concentration. Monthly means also cannot serve episode-scale exposure work. Still, boundary-layer height is the dominant meteorological term in almost every surface PM model, and a validated European product that beats reanalysis is directly usable as a covariate.

doi: 10.5194/amt-19-5697-2026

**TROPOMI SO2 EAKF assimilation 2026** Atmosphere

*Assimilating TROPOMI $SO_2$ columns into a regional chemical transport model*

An Ensemble Adjustment Kalman Filter implemented in the MINNI regional CTM through DART, assimilating Sentinel-5P TROPOMI $SO_2$-COBRA total columns over continental Europe for August 2023, with a 20-member ensemble perturbing emissions and boundary conditions. The filter constrained ensemble variance around power-plant and volcanic plumes; the monthly domain-mean column correction was **$2\times10^{-5}$ mol m$^{-2}$** with localised maxima of **$3.3\times10^{-4}$**, and near-surface concentration corrections reached **2.6 $\mu$g m$^{-3}$**, validated against in-situ data. The stated limitations are real ones: small ensemble, static vertical localisation, and the usual temporal fading of initial-condition corrections. Relevance here is indirect but concrete — $SO_2$ is a sulfate precursor, and column-to-surface transfer skill is the same problem satellite-based $PM_{2.5}$ products have to solve.

doi: 10.3390/atmos17090875

**SE India monsoon aerosol 2026** Atmosphere

*Seasonal aerosol optical properties and radiative effect at a rural Indian receptor site*

MICROTOPS-II sunphotometry from April 2021 to December 2023 at a rural site in Southeast India, with trajectory-based source attribution (PSCF, CWT) and OPAC–SBDART radiative transfer. Annual mean **$AOD_{500}$ 0.56 $\pm$ 0.22**, peaking pre-monsoon (0.66 $\pm$ 0.19) and in winter (0.64 $\pm$ 0.23) and falling to 0.49 $\pm$ 0.21 in the monsoon. **Angstrom exponent 1.30** $\pm$ **0.24** in the dry seasons against **0.83** $\pm$ **0.37** in the monsoon marks the fine-to-coarse shift, corroborated by negative spectral curvature. Precipitable water rose 2.00 to 4.35 cm winter to monsoon, which confounds the optical size retrieval through hygroscopic growth — the paper says so. Column optical depth is not surface mass, and a single rural receptor cannot be generalised; its value is as a South Asian background record against which urban surface networks can be referenced.

doi: 10.3390/atmos17090876

**PPDRTL leaf-image AQI 2026** PLoS One

*Predicting roadside air quality from image texture of pollutant deposits on tree leaves*

High-resolution images of roadside leaves at high, medium and low traffic-density sites in Haryana, segmented with PSO, DPSO and FODPSO, reduced to contrast, entropy and standard deviation features and regressed with ordinary linear regression onto Haryana State Pollution Control Board reference AQI. The FODPSO variant reaches $R^2$ **up to 0.894** with **RMSE 8.76–12.34 $\mu$g m$^{-3}$** and reported site accuracies of 88.9–91.4%. The inference is weak for a reason worth naming precisely: the **pilot dataset is ten days**, no held-out test set is described, and leaf deposition is a *cumulative* integrator being regressed onto an *instantaneous* daily index, so the model is fitting a mismatched target. Mixing an $R^2$ against a unitless AQI with an RMSE in $\mu$g m$^{-3}$ compounds the confusion. Filed as a proxy-sensing curiosity, not a method.

doi: 10.1371/journal.pone.0326588

**SIBS inexpensive single-particle 2026** Aerosol Sci Technol

*METADATA ONLY — feasibility of a low-cost single-particle SIBS instrument*

**No abstract existed in Crossref, Europe PMC, Semantic Scholar or on the publisher page on the entry date**, and Taylor & Francis refused the fetch. What the record establishes: a feasibility study for an *inexpensive* single-particle spark-induced-breakdown-spectroscopy instrument has been published in the field's instrumentation journal of record. What it does not establish: any detection limit, element set, particle-size range, spark energy, hit rate or cost figure — none of which can be inferred from the title, and none of which is asserted here. This is the most directly relevant record of the day to low-cost elemental speciation and is flagged for retrieval once the abstract deposits.

doi: 10.1080/02786826.2026.2721653

**3D-LSS particle shape 2026** J Aerosol Sci

*METADATA ONLY — enhanced 3D light-scattering sensor for online particle shape*

**No abstract retrievable on the entry date.** The title commits to two things: an *enhanced* three-dimensional light-scattering sensor for online aerosol shape characterisation, and a validation against test aerosols. Neither the size range, the angular sampling scheme, the sphericity metric nor the validation particle set is recoverable, and none is asserted here. Shape is the property optical low-cost sensors silently assume away when they convert scattering to mass, so an online shape channel is directly relevant; the predecessor 3D-LSS work established the concept at 500 nm–5 $\mu$m, and what "enhanced" means against that baseline is precisely what the abstract would settle.

doi: 10.1016/j.jaerosci.2026.106860

**BC coating optics 2026** J Aerosol Sci

*METADATA ONLY — metal-oxide coatings and the optical properties of black carbon*

**No abstract retrievable on the entry date.** The record establishes that the optical effect of $V_2O_5$, NiO and $Fe_2O_3$ surface coatings on black carbon aggregates has been characterised; it establishes no absorption enhancement factor, wavelength dependence, coating thickness or morphology treatment, and none is asserted here. The reason to track it: absorption-based BC measurement — aethalometers, photoacoustic instruments and the filter-based methods most networks run — is systematically biased by the coating enhancement factor, and the metal-oxide coatings specifically are the industrial and brake-wear case rather than the usual organic or sulfate one.

doi: 10.1016/j.jaerosci.2026.106885

## Exposure assessment & modelling 4 records

**Accra traffic intersections 2026** Environ Monit Assess

*Meteorological drivers of $PM_{2.5}$ at nine Accra traffic hotspots*

An Aeroqual Ranger portable monitor at nine traffic intersections in Accra across the harmattan and non-harmattan seasons, against Ghana Meteorological Agency station data. Median $PM_{2.5}$ was **54.90 $\mu$g m$^{-3}$** at high-density sites, **34.26** at commercial and **16.75** at low-density — the cleanest category still exceeds the WHO 24 h guideline of 15. Temperature correlated positively ($r = 0.353$, $p < 0.01$); rainfall, wind speed and humidity negatively outside the harmattan, while in the harmattan wind speed flipped weakly positive ($r = 0.079$, $p = 0.450$, i.e. null) consistent with dust resuspension. The analysis is Pearson correlation only, with no multivariable adjustment and no traffic counts, so "the dominant factor remains localised vehicular emissions" is a residual argument rather than a measured one. A single portable instrument across nine sites also gives no co-location error term.

doi: 10.1007/s10661-026-15889-8

**Xiamen chlorine NO2 uptake 2026** Atmos Chem Phys

*Chlorine-enhanced nocturnal $NO_2$ uptake and its nitrate yield in a coastal city*

Field observations in coastal Xiamen combined with a machine-learning attribution and a multiphase box model, contrasting sea-land-breeze nights against non-SLB nights. HONO and particulate nitrate both rose on SLB nights, with a mean $NO_2$ uptake rate constant of $k_{NO_2} = 9.70 \times 10^{-6}$ **s$^{-1}$**; chlorine ranked as the single most important driver of the enhancement. Feeding that constant back into the box model resolved the standing underestimation of both HONO and nitrate production, with nocturnal uptake accounting for **83.9% of HONO** and **46.4% of nitrate formation**. The machine-learning importance ranking is associational and cannot separate chlorine from the marine air mass that carries it, and one coastal city in one season limits transfer. Relevance is to secondary inorganic PM: nitrate is a large and poorly forecast fraction of coastal urban $PM_{2.5}$.

doi: 10.5194/acp-26-12715-2026

**Brisbane Aitken-mode PNC 2026** Environ Int

*METADATA ONLY — long-term Aitken-mode particle number trends in Brisbane*

**No abstract retrievable on the entry date** despite *Environ Int* being a PubMed-indexed journal — the DOI indexed in Crossref before either the abstract or the PubMed record deposited. The title commits to a long-term change in Aitken-mode particle number concentration attributed across meteorology, traffic and biomass burning. Neither the record length, the trend sign or magnitude, the instrument, nor the attribution method is recoverable, and none is asserted here. Long records of sub-100 nm number concentration are rare and this one would bear directly on whether mass-based networks are tracking the health-relevant fraction; flagged for retrieval.

doi: 10.1016/j.envint.2026.110514

**W Sub-Saharan Africa PM monitoring 2026** Atmos Pollut Res

*METADATA ONLY — $PM_{2.5}$ and $PM_{10}$ monitoring capacity in Western Sub-Saharan Africa*

**No abstract retrievable on the entry date.** The record establishes that a challenges-and-perspectives treatment of $PM_{2.5}$ and $PM_{10}$ monitoring and estimation across Western Sub-Saharan Africa has been published; whether it is a systematic review, a narrative synthesis or a data-driven assessment cannot be determined from the metadata, and no finding is asserted here. Western Sub-Saharan Africa has the sparsest reference network of any populated region and is the canonical justification for low-cost sensor deployment, so a current capacity assessment is worth retrieving — particularly alongside the Accra hotspot record in this same issue.

doi: 10.1016/j.apr.2026.103195

## Occupational & indoor 2 records

**Oklahoma cannabis facility IAQ 2026** Ann Work Expo Health

*Seven-instrument occupational air-quality characterisation of a cannabis grow room*

Continuous monitoring from 10 July to 1 August 2023 in a 186 m$^2$ HVAC-equipped medical cannabis cultivation room holding roughly 900 plants on a 12:12 photoperiod. Seven instruments across three spatial zones at 1.5 m breathing height — three Aranet4 NDIR $CO_2$ sensors, two TSI AirAssure multi-parameter monitors carrying size-resolved PM, and two PurpleAir PA-II-FLEX dual-channel counters — produced ~**113,000 quality-controlled records**. $CO_2$ tracked the light cycle, with means of **979–1,086 ppm** and maxima above **2,200**; nighttime elevation is attributed to plant metabolism and enrichment rather than worker respiration, since workers leave at lights-out. **Median $PM_{2.5}$ was low at 1.4–3.1 $\mu$g m$^{-3}$** with episodic activity-driven spikes. One facility, one room, one summer month, and no personal sampling, so this is a determinants pilot rather than an exposure assessment — but it is the rare occupational study that runs consumer-grade and research-grade particle instruments side by side.

doi: 10.1093/annweh/wxag074

**Ethiopia dormitory ventilation 2026** Indoor Environments

*METADATA ONLY — occupant-driven ventilation in naturally ventilated dormitories*

**No abstract retrievable on the entry date.** The title commits to a field study of occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories. The pollutants measured, the ventilation-rate method (tracer gas, $CO_2$ decay or modelled), the number of rooms and the season are all unrecoverable and none is asserted here. Occupant-controlled natural ventilation is the largest uncontrolled term in indoor exposure models for warm low-resource settings, and measured air-change data from Sub-Saharan Africa are scarce enough that the study is worth retrieving on those grounds alone.

doi: 10.1016/j.indenv.2026.100192

## Other clinical endpoints 1 record

**Multimorbidity meta-analysis 2026** BMJ Public Health

*Pooled risk of multimorbidity per 5 $\mu g\,m^{-3}$ of PM and $NO_2$*

Systematic review to 1 April 2026 across four databases (PROSPERO CRD42024613048); 22 studies included, 21 meta-analysed with random effects. Pooled adjusted risk ratios per 5 $\mu g\,m^{-3}$: **$PM_{2.5}$ 1.15 (1.08–1.23)** from 4 studies at $I^2 = 93.0\%$; **$PM_{10}$ 1.27 (1.11–1.47)** from 3 at $I^2 = 99.5\%$; **$PM_1$ 1.07 (1.05–1.09)** from 2 at $I^2 = 0.0\%$; **$NO_2$ 1.12 (1.00–1.26)** from 4 at $I^2 = 96.6\%$. Nitrogen oxides, black carbon, $SO_2$, ozone and CO showed nothing. The limitation is structural, not incidental: harmonising to a per-5 $\mu g\,m^{-3}$ increment discarded most of the 22 studies, leaving two to four per pollutant, and at $I^2 = 93$–$99.5\%$ the $PM_{2.5}$ and $PM_{10}$ pooled values are averages over non-exchangeable studies. **Zero eligible studies** on polypharmacy — half the stated aim — or on household air pollution, older adults, the Americas or Oceania.

doi: 10.1136/bmjph-2025-004782

## Reproductive & developmental 1 record

**France SUDI case-crossover 2026** Paediatr Perinat Epidemiol

*Short-term air pollution and sudden unexpected death in infancy, national registry*

**1,078 SUDI cases** from the French national registry, 2015–2022, with modelled daily $PM_{10}$, $PM_{2.5}$, $NO_2$ and $O_3$ assigned at each infant's residence. A time-stratified case-crossover design with conditional logistic regression, adjusted for public holidays, influenza-like illness and daily mean temperature, examined lags to six days. **PM exposure was consistently associated with higher odds of SUDI at lags 3–5 and across the last week of life**, strongest in cumulative and multi-pollutant models; $NO_2$ showed a trend only, and effects were larger in spring. The case-crossover design self-matches on fixed confounders, which is its strength, but the abstract reports no point estimates or intervals — so the association's magnitude cannot be judged and this record contributes nothing to the forest plot. Residence-level modelled exposure also ignores where infants actually sleep and travel. SUDI remains an aetiologically open outcome; a short-term environmental trigger is a plausible and previously little-tested component.

doi: 10.1111/ppe.70196

## Burden, policy & mitigation 2 records

**Puyang PM2.5-O3 decade 2026** Preprints.org

*A decade of diverging $PM_{2.5}$ and ozone in a Chinese oilfield city*

**Preprint, not peer reviewed.** Regulatory monitoring records for Puyang, one of the "2+26" channel cities, 2015–2025. **$PM_{2.5}$ fell 79.6 to 39.3 $\mu g\,m^{-3}$ while MDA8 ozone rose 123.0 to 189.6** — the canonical Chinese divergence, here in a city whose oil and gas VOC inventory makes it an unusually clean test of it. A positive $PM_{2.5}$–$O_3$ correlation appeared only in the compound-pollution corner ($O_3 > 160$, $PM_{2.5} < 35$), with seasonal $r$ of 0.38 winter, 0.37 summer, 0.26 autumn, 0.21 spring — all weak. Ozone production efficiency declined over the record, correlated with $PM_{2.5}$ at $R = 0.46$–$0.65$, and stayed above 7 monthly, which the authors read as NOx-limited. OPE inferred from ambient monitoring rather than measured, no VOC speciation reported, and a single city; the control recommendation rests on that inference.

doi: 10.20944/preprints202609.0349.v1

**Australian air health risk index 2026** Environ Pollut

*METADATA ONLY — community-specific weather and air health risk indices*

**No abstract retrievable on the entry date**, despite *Environ Pollut* being PubMed-indexed — the same Crossref-ahead-of-deposit lag seen elsewhere in this issue. The title commits to community- and outcome-specific weather-and-air health risk indices built with explainable machine learning across **66 Australian communities**. The pollutant set, the health outcomes, the index construction and its skill are all unrecoverable and none is asserted here. Outcome-specific and community-specific index construction is the methodological direction that would replace one-size AQI banding, and 66 communities is enough to test whether the community-specific part earns its complexity; flagged for retrieval.

doi: 10.1016/j.envpol.2026.129117

# Corpus register

| First author | Focus | Journal | Design | Metric | Region |
|---|---|---|---|---|---|
| **Reproductive & developmental** | | | | | |
| France SUDI case-crossover | Short-Term Effect of Air Pollutants on Sudden Unexpected Death in Infancy: A Case-Crossover Study Based on the French National Registry | *Paediatr Perinat Epidemiol* | Time-stratified case-crossover | Modelled residential daily $PM_{10}$, $PM_{2.5}$, NO2, O3 | France |
| **Occupational & indoor** | | | | | |
| Ethiopia dormitory ventilation | Occupant-driven ventilation and indoor exposure dynamics in naturally ventilated Ethiopian university dormitories | *Indoor Environments* | Metadata only (no abstract) | Indoor exposure under occupant-driven natural ventilation | Ethiopia |
| Oklahoma cannabis facility IAQ | Multi-parameter characterization of indoor air quality in a medical marijuana cultivation facility in Oklahoma: a pilot study of occupational exposure determinants | *Ann Work Expo Health* | Multi-instrument field evaluation | Size-resolved PM by TSI AirAssure + 2 PurpleAir PA-II-FLEX | United States |
| **Sensing, forecasting & instrumentation** | | | | | |
| 3D-LSS particle shape | Enhanced three-dimensional light scattering sensor (3D-LSS) for online characterization of aerosol particle shape: Development and validation with test aerosols | *J Aerosol Sci* | Metadata only (no abstract) | Angularly resolved elastic scattering, single particle | Not stated |
| BC coating optics | Effects of V2O5, NiO, and Fe2O3 surface coatings on the optical properties of black carbon aggregates | *J Aerosol Sci* | Metadata only (no abstract) | Optical properties of coated BC aggregates | Laboratory |

| First author | Focus | Journal | Design | Metric | Region |
|---|---|---|---|---|---|
| Europe BLH regression kriging | Modelling monthly Boundary Layer Height maps combining radiosonde, satellite, and reanalysis over Europe | *Atmos Meas Tech* | Regression kriging map product | Gridded monthly PBL height, 25 km, 00 and 12 UTC | Europe |
| PPDRTL leaf-image AQI | PPDRTL: A novel framework for predicting pollutant deposition on roadside tree leaves using linear regression | *PLoS One* | Image-feature regression proxy | AQI regressed on leaf-surface image texture features | India |
| SE India monsoon aerosol | Monsoon-Regulated Aerosol Variability and Radiative Effects over a Rural Receptor Site in Southeast India | *Atmosphere* | Long-term field measurement campaign | MICROTOPS-II sunphotometer AOD500 and Angstrom exponent | India |
| SIBS inexpensive single-particle | Feasibility of an inexpensive single-particle SIBS instrument | *Aerosol Sci Technol* | Metadata only (no abstract) | Single-particle spark-induced breakdown spectroscopy | Not stated |
| TROPOMI SO2 EAKF assimilation | Assimilation of SO2 TROPOMI Retrievals at the European Scale with EAKF Implemented in MINNI Through DART | *Atmosphere* | Satellite + chemical transport model | TROPOMI SO2-COBRA total column into MINNI CTM | Europe |
| W Sub-Saharan Africa PM monitoring | Monitoring and estimating $PM_{2.5}$ and $PM_{10}$ concentrations in Western Sub-Saharan Africa: Challenges and perspectives | *Atmos Pollut Res* | Metadata only (no abstract) | Regional $PM_{2.5}$/$PM_{10}$ monitoring and estimation capacity | West Africa |
| **Exposure assessment & modelling** | | | | | |
| Accra traffic intersections | Meteorological drivers of ambient $PM_{2.5}$ pollution at urban traffic intersections in Accra, Ghana | *Environ Monit Assess* | Portable sensor monitoring campaign | Aeroqual Ranger portable dust monitor, 9 traffic hotspots | Ghana |
| Brisbane Aitken-mode PNC | Long-term changes in Aitken-mode particle number concentration in Brisbane: roles of meteorology, traffic, and biomass burning | *Environ Int* | Metadata only (no abstract) | Long-term Aitken-mode particle number concentration | Australia |
| N China indoor Finf 48 homes | Multivariable Determinants of Indoor $PM_{2.5}$ and Infiltration Factors in 48 Residential Buildings Across Eight Northern Chinese Cities: A Seasonal Monitoring and Linear Mixed-Effects Modelling Study | *Atmosphere* | Seasonal indoor-outdoor monitoring + LMM | Paired indoor/outdoor $PM_{2.5}$ at 5 min, hourly aggregate | China |
| Xiamen chlorine NO2 uptake | Chlorine enhances nocturnal heterogeneous uptake of NO2 in coastal atmosphere under sea-land breeze circulation | *Atmos Chem Phys* | Kinetic model + field validation | Nocturnal particulate nitrate and HONO under sea-land breeze | China |
| **Burden, policy & mitigation** | | | | | |
| Australian air health risk index | Constructing Community- and Outcome-specific Weather and Air Health Risk Indices Using Explainable Machine Learning: A Multicentre Study in 66 Australian Communities | *Environ Pollut* | Metadata only (no abstract) | Community- and outcome-specific air health risk index | Australia |
| Puyang $PM_{2.5}$-O3 decade | Long-Term Trends and Nonlinear Interactions of $PM_{2.5}$-O3 Compound Pollution in a Typical Oilfield City of the 2+26 Region | *Preprints.org* | Monitoring-record analysis | Regulatory $PM_{2.5}$ and MDA8 O3, Puyang, 2015-2025 | China |
| **Other clinical endpoints** | | | | | |
| Multimorbidity meta-analysis | Association between air pollution, multimorbidity and polypharmacy: a systematic review and meta-analysis | *BMJ Public Health* | Systematic review + random-effects meta-analysis | Long-term ambient $PM_{2.5}$/$PM_{10}$/PM1/NO2, per 5 ug m-3 | Global |

### Provenance and method

Entry window **8 September 2026**, a single day, run against `[Date - Entry]`. The run began at 22:00 local, satisfying the standing rule that the current calendar day is not opened before then; `last_entry_date` was 7 September, so exactly one day was owed and no gap exists. **PubMed** (NCBI E-utilities) returned **19 health-axis** and **3 sensing-axis** hits for 20 unique records. The **PubMed connector**, run independently on both axes, returned **19** and **2** — the identical health set and a subset of the sensing set, so it added nothing; this is the first run in the archive where it is pure confirmation. Seven of the 20 PubMed records had already shipped in earlier issues and were dropped on PMID/DOI. **Europe PMC** (`CREATION_DATE`) returned **1** raw, a Preprints.org record neither PubMed leg carried, and it shipped. **Crossref-by-ISSN** returned **59 raw** and shipped **12** — two thirds of the issue, from *Aerosol Sci Technol*, *J Aerosol Sci*, *Atmos Meas Tech*, *Atmos Chem Phys*, *Atmosphere*, *Atmos Pollut Res*, *Indoor Environments*, *Environ Int* and *Environ Pollut*. **arXiv** and **OpenAlex** both returned HTTP 429 and contributed nothing; OpenAlex's created-date filter remains paywalled independently of the rate limit. **Consensus** returned three hits, **all three already published here** on 1, 2 and 5 August — verified by resolving each title to a DOI and querying `seen.json`, not by inspecting publication dates. **Scholar Gateway** was queried once, for the SIBS instrument paper; its corpus is Wiley-weighted and held no same-day Taylor & Francis or Elsevier deposit, so it is not a usable abstract source for records this fresh. **ClinicalTrials.gov** posted **1,135 updates registry-wide** on 8 September and **zero** matching air pollution or particulate matter — a topical zero confirmed by no-term control, unlike yesterday's federal-holiday zero. `trials.json` unchanged at 11. **Thirty-four records were excluded with a logged reason**: ten duplicates already in `seen.json`, and beyond those, ozone-only and greenhouse-gas endpoints, cigarette-smoke toxicology, aquatic and materials chemistry, non-aerosol metrology, a correction notice, and two records whose particulate scope could not be confirmed without an abstract. **Seven in-scope records carry no abstract anywhere** — checked per DOI against Crossref, Europe PMC, Semantic Scholar and the publisher page — and ship as `Metadata only`, tier C. Their entries state what the metadata establishes and assert no finding; treat them as pointers, and note that the corresponding rows of the register carry no result. The DOI gate passed **18 of 18 with zero warnings**. Three new design labels were added to the plotting maps in the same edit as the records that use them; the geography map fired no unmapped diagnostic. *Global / multi-region* remains a fallback bucket that absorbs unstated settings and bench work — here 2 of its 4 are metadata-only records with no site. Subtopic, design, particle-metric and geography labels are assigned from title, abstract and keywords; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim; all four come from one meta-analysis — see the forest-plot caption. All DOI links resolve to the publisher of record. 